Analysis of Healthcare Data to Identify Trends and Patterns

1. Project objective:

This project aims to analyse **healthcare** data to identify key **trends** in **patient admissions**, **hospitals**, **medical conditions**, and billing from recent years **2022, 2023** and **2024**. It includes data cleaning, data transformation and data modeling and building an interactive dashboard for **deeper analyse** and also to analyse the impact of **admission types**, **insurance providers** on overall **healthcare costs** and **patient outcomes**. Finally to provide actionable insights that help **improve hospital efficiency**, **optimize resource utilization** and **cost optimization** for **decision-making**.

2. Data Source:

Source: https://huggingface.co/datasets/Kilos1/healthcare_dataset/tree/main

Time Line: 2019 – 2024

Domain: Healthcare

3. Tools Used:

- **Microsoft Excel**
- **Microsoft Power Query**
- **Microsoft Power BI**

4. Project Methodology and Implementation:

- **Data Cleaning**
- **Feature Creation using Excel, Power Query, and Power BI**
- **Data Modeling**
- **Objectives & Findings**
- **Dashboard & Insights**

5. Attribute (Column/Features) Details:

Attribute Name	Description	Data Type
Patient ID	Unique identifier assigned to each patient	Integer
Age	Age of the patient in years	Integer
Age Group	Categorized age group of the patient	Text
Gender	Gender of the patient	Text
Blood Type	Blood group of the patient	Text
Medical Condition	Primary diagnosis or health condition	Text
Hospital ID	Unique ID assigned to each hospital	Text
Hospital	Name of the hospital where patient was treated	Text
Insurance Provider	Insurance company covering the patient	Text

Billing Amount	Total medical bill charged for treatment	Decimal (Currency)
Billing Category	Category based on billing amount (Low/Medium/High)	Text
Admission ID	Unique identifier for each admission record	Text
Date of Admission	Date when patient was admitted	Date
Admission Type	Type of admission (Emergency, Urgent, Elective)	Text
Discharge Date	Date when patient was discharged	Date
Length of Stay	Total number of days patient stayed in hospital	Integer
Stay Category	Category based on length of stay	Text
Medication	Medicine prescribed to the patient	Text
Test Results	Result of medical tests conducted	Text

6. Data Cleaning steps involved:

Data cleaning was performed to improve **data quality**, **ensure consistency**, and enhance the **reliability of analysis**. The following steps were carried out:

- Checked for **Duplicates** in **Power Query** and no duplicate records were found, ensuring data **uniqueness** and there is no **empty cells** were found in each column.
- Removed **Patient Name** column because it is not useful for **analysis** because every name is **unique** and no **grouping** is possible and it doesn't give **insights** about **health**, **costs** or **trends** particularly in this project.
- Removed **Doctor Name** column as it was not relevant for **analytical insights** and to simplify the dataset.
- Filtered **Medical Conditions** to include only **Cancer**, **Diabetes**, and **Obesity** focusing only the analysis on major health conditions.
- Filtered Year to include only data from **2022**, **2023**, and **2024** to ensure **recent** and **relevant** insights.
- Removed **negative values** in **Billing Amount** to avoid incorrect **financial calculations** and maintain **data accuracy**.
- Removed **Room Number** column as it did not contribute to **meaningful analysis**.
- Filtered **Test Results** to include only **Normal** and **Abnormal**, removing **Inconclusive** values for **clearer** interpretation.
- Handled **Data Types** by correcting and standardizing data types in both **Excel** and **Power BI** to ensure proper **aggregation** and **calculations**.

Applied **Transformations** in **Power Query**, including **capitalization** of text values, **trimming** extra spaces, and **replacing** inconsistent values to improve data consistency. These

steps ensured that the dataset was **clean**, **consistent**, and ready for **accurate** analysis and **visualization**.

7. Feature Creation using Excel, Power Query, and Power BI:

7.1 Created additional Columns as **Patient ID**, **Hospital ID** and **Admission ID** using **excel** for **deeper** analysis.

Purpose:

- To uniquely identify records across **different** tables.
- To support proper relationships in the **data model** (Star Schema).
- To avoid **ambiguity** and duplication during analysis.

7.2 Created a column in **power query** Named as **Length of Stay** by selecting **add column → custom Column**.

Purpose:

- To calculate the number of days a patient stayed in the hospital.
- Helps in analyzing hospital efficiency and patient flow.

7.3 Created a column Named as **Age Group** for age group categorization by selecting **add column → conditional column**.

Purpose:

- Enables demographic segmentation.
- Helps identify which age group is most affected by specific conditions.

7.4 Created a column Named as **Billing Category** for billing categorization by selecting **add column → custom column**.

Purpose:

- Groups patients based on treatment cost.
- Useful for revenue analysis and cost distribution.

7.5 Created a column Named as **Stay Category** for stay categorization by selecting **add column → custom column**.

Purpose:

- Helps analyze **hospitalization** patterns.
- Useful for **operational** efficiency and **resource** planning.

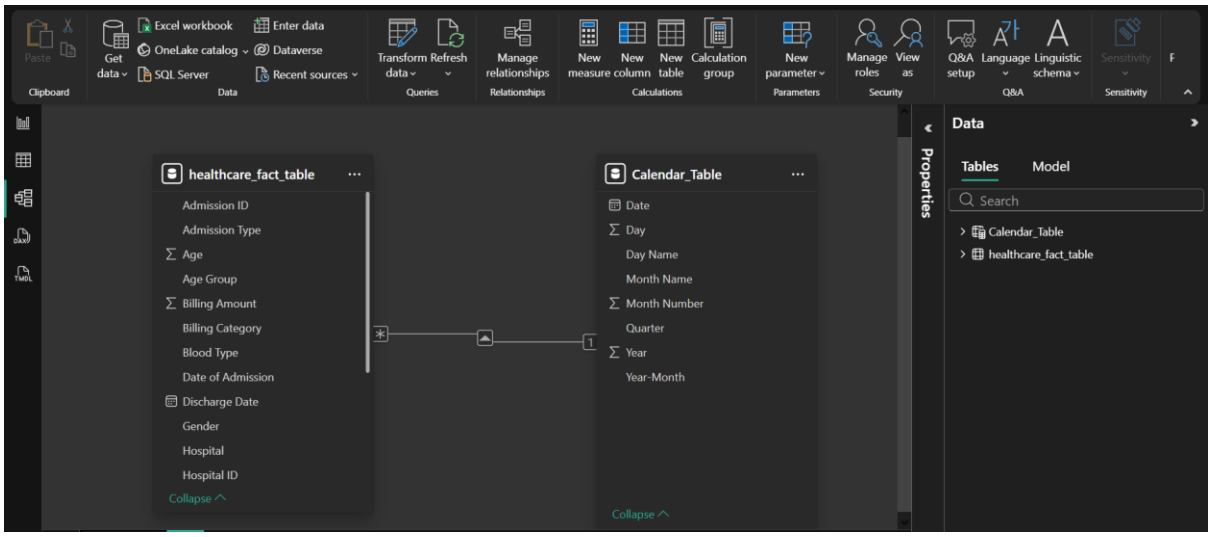
7.6 A dedicated **Calendar Table** was created using **Time DAX** to support **time-based** analysis and improve **data modeling**.

Generated columns are **Year**, **Month Number**, **Month Name**, **Year-Month**, **Quarter & Day Name**.

Purpose:

- Enables time intelligence calculations.
- Supports trend analysis (monthly, quarterly, yearly).
- Improves performance and flexibility of reports.

8. Data Modeling:



- The dataset was structured using a **Star Schema** approach to ensure efficient data analysis and reporting in Power BI. The central fact table (**healthcare_fact_table**) contains transactional data such as **patient admissions**, **billing amount**, **medical details**, and **hospital information**.
- A separate calendar **dimension table** (**Calendar_Table**) was created using DAX to enable **time-based analysis**. This table includes attributes such as **Year**, **Month**, **Quarter**, **Day**, and **Year-Month**, which support trend analysis and time intelligence functions.
- A one-to-many relationship was established between the Calendar Table and the Fact Table using the date field, where the Calendar Table acts as the dimension and the Fact Table as the transactional data source.

This model improves:

- Query performance
- Data organization
- Time-based analysis (monthly, yearly trends)
- Scalability for future dimensions

9. Objectives & Findings For KPI:

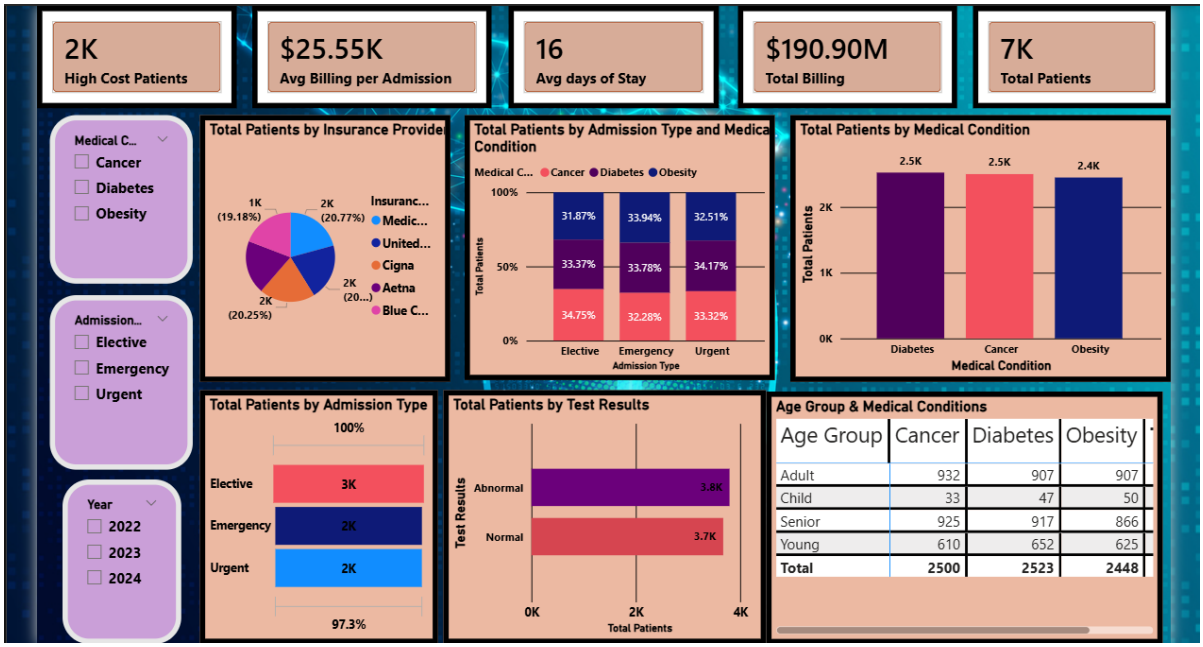
- Analyzed **Total patients by** using **card visual** as KPI and the dataset contains approximately **7K unique patients**, indicating overall hospital volume and demand.
- Evaluated **Total billing** by using **card visual** as KPI and the total **revenue** generated is approximately **$190.90M**, indicating overall hospital financial performance.
- Analyzed a **Total admissions** using as measure and the found that the total admission is **7k admissions** and also analyzed the **average billing per admission** by using **card**

visual as KPI which Indicates average cost per visit, helping understand pricing efficiency and revenue per admission and the average cost per admission is **$25.55k.**

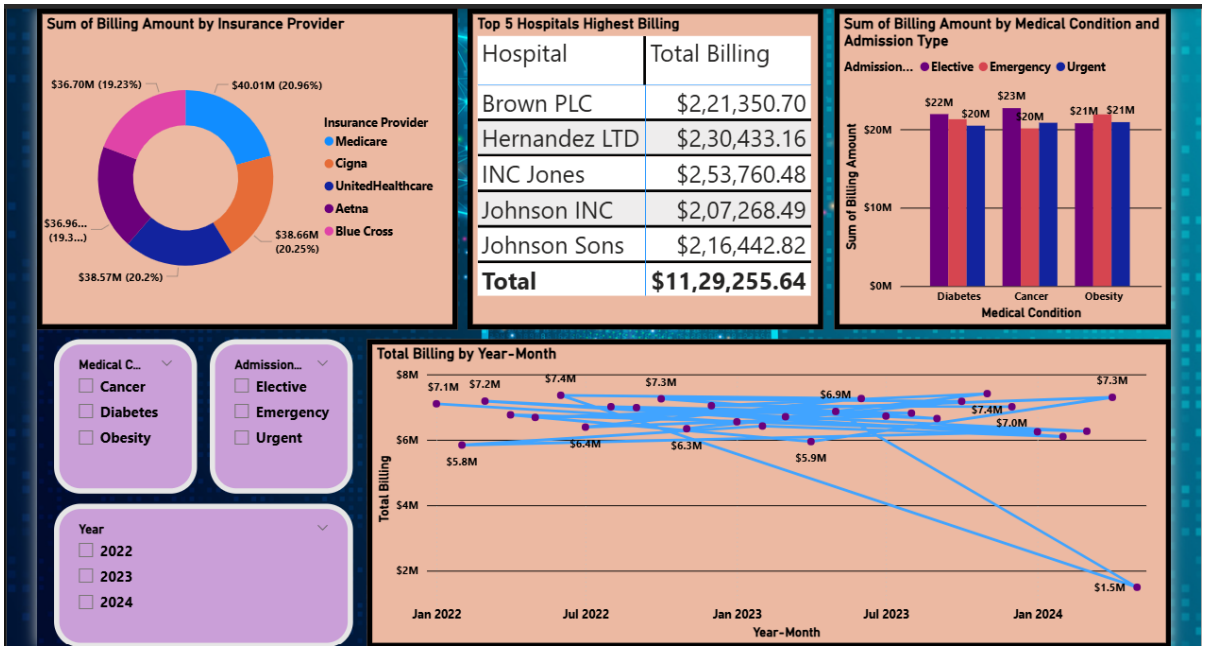
- Analyzed **hospital efficiency** by using **Card visual** and the average stay is **16 days** this one is indicating how long patients remain admitted and helping evaluate resource utilization.
- Identified **high-cost patient** using **Card visual** and found approximately **2K patients** fall under high billing category, indicating a significant contribution to overall hospital revenue.

10. Dashboard & Insights:

Patient & Admission Analysis



Billing & Financial Analysis



10.1 Chart Name: Total Patients by Insurance Provider & Chart Type: Pie Chart

This chart illustrates the distribution of **total patients** across different **insurance providers** to understand their contribution. **Medicare** has the **highest** share with **1,552** patients **(20.77%)**, followed by **UnitedHealthcare** with **1,523 (20.39%)** and **Cigna** with **1,513 (20.25%)**. **Aetna** accounts for **1,450** patients **(19.41%)**, while **Blue Cross** has the **lowest** with **1,433 (19.18%)**. The near-equal distribution indicates that patient coverage is **evenly** spread across providers. This insight highlights that no single insurance provider **dominates**, ensuring **balanced revenue** and patient dependency.

10.2 Chart Name: Total Patients by Admission Type and Medical Condition & Chart Type: 100% Stacked Column Chart

This chart shows the percentage distribution of **medical conditions** (Cancer, Diabetes, Obesity) across different **admission types**. It is used to compare how diseases are **proportionally spread** within **Elective**, **Emergency**, and **Urgent admissions**. In Elective cases, Cancer accounts for **34.75%** which is **highest**, Diabetes **33.37%**, and Obesity **31.87%**, while Emergency cases show a distribution with Cancer **32.28%**, Diabetes **33.78%** and Obesity **33.94%** which is **highest**. Urgent admissions show a distribution with Cancer **33.32%**, Diabetes **34.17%** which is **highest** and Obesity **32.51%**. The overall insight indicates that all three conditions are almost **equally** distributed across admission types, showing no strong **dominance** of any condition.

10.3 Chart Name: Total Patients by Medical Condition & Chart Type: Clustered Column Chart

This chart compares the **total number of patients** across different **medical conditions** to identify the most prevalent disease. **Diabetes** has the **highest** number of patients with **2,523**, followed closely by **Cancer** with **2,500**, while **Obesity** has **2,448** patients. The difference between conditions is **minimal**, indicating a nearly **balanced** distribution of diseases. This suggests that all three conditions contribute almost equally to the overall patient population.

10.4 Chart Name: Total Patients by Admission Type Chart Type: Funnel Chart

This chart represents the distribution of patients across different **admission types** to analyze **patient flow**. **Elective** admissions have the **highest** count with **2,535** patients, followed by Emergency with **2,469** and Urgent with **2,467**. The values are very **close**, indicating a nearly uniform distribution among all admission types. This suggests that patient inflow is **balanced**, with a slightly higher preference for **elective** admissions.

10.5 Chart Name: Total Patients by Test Results & Chart Type: Clustered Bar Chart

This chart compares the number of patients based on **test results** to evaluate overall health outcomes. **Abnormal** results account for **3,796** patients, slightly **higher** than **Normal** results with **3,675** patients. This suggests a slightly **higher occurrence of health issues** among patients, highlighting the need for closer medical attention.

10.6 Chart Name: Age Group & Medical Conditions & Chart Type: Matrix Table

This chart provides a detailed comparison of **medical conditions** across **different age groups** to identify **demographic trends**. **Adults** show the **highest** cases with **932** (**Cancer**), **907** (**Diabetes**), and **907** (**Obesity**), followed by **Seniors** with **925** (**Cancer**), 917 (**Diabetes)**, and 866 (**Obesity**). **Young** patients with **610** (**Cancer**), **652** (**Diabetes**), and **625** (**Obesity**). **Children** have the **lowest** cases with **33** (**Cancer**), **47** (**Diabetes**), and **50** (**Obesity**). This indicates that disease prevalence is significantly higher in Adult and Senior groups compared to Young and Child populations.

10.7 Chart Name: Total Billing by Insurance Provider & Chart Type: Donut Chart

This chart shows the distribution of **total billing amount** across different **insurance providers** to analyze **revenue** contribution. **Medicare** generates the **highest** billing with **$40.01M** (20.96%), followed by **Cigna** with **$38.66M** (20.25%) and **UnitedHealthcare** with **$38.57M** (20.2%). **Aetna** contributes **$36.96M** (19.3%), while **Blue Cross** has the **lowest** at **$36.70M** (19.23%). The near-equal percentages indicate **balanced** revenue contribution with no single provider **dominating**.

10.8 Chart Name: Top 5 Hospitals by Highest Billing & Chart Type: Table

This chart displays the **top 5 hospitals** based on **total billing amount** to identify **high-performing institutions**. **INC Jones** has the **highest** billing with **$2,53,760.48**, followed by **Hernandez LTD** with **$2,30,433.16** and **Brown PLC** with **$2,21,350.70**. **Johnson Sons** with **$2,16,442.82** and **Johnson INC** with **$2,07,268.49**. The total billing from these top hospitals amounts to **$11,29,255.64**, highlighting their significant contribution to **overall revenue**.

10.9 Chart Name: Billing Amount by Medical Condition and Admission Type Chart Type: Clustered Column Chart

This chart compares **total billing** across **medical conditions** and **admission types** to analyze cost variations. For **Diabetes**, billing is **$21,935,914.37** (Elective), **$21,281,963.10** (Emergency), and **$20,459,448.01** (Urgent), while **Cancer** shows **$22,697,798.39** (Elective), **$20,111,166.94** (Emergency), and **$20,831,173.69** (Urgent). **Obesity** records **$20,768,930.30** (Elective), **$21,903,099.45** (Emergency), and **$20,912,327.63** (Urgent). The key insight is that Elective admissions generally generate higher billing, with **Obesity** showing a peak in **Emergency** cases.

10.10 Chart Name: Total Billing by Year-Month Chart Type: Line Chart

This chart shows the **trend** of **total billing over time** to analyze **financial performance** across **year-months**. Billing generally ranges between **$5.8M** and **$7.4M**, with **peaks** around **$7.4M** (**2022**) and **$7.3M** (**2024**). However, there is a significant drop to **$1.5M** in **2024**, indicating an **unusual** decline. The key insight is that while billing remains mostly **stable**, the sharp **drop** in **2024** may indicate **incomplete** data or a sudden **decrease** in activity.

11. Recommendations And Actionable Insights:

These recommendations are intended for **hospital management** and **healthcare decision-makers** to improve **operational** efficiency, **patient** management, and **financial** performance.

- The data shows that performance in **2022** and **2023** was **stable**, so existing strategies should be continued.
- Focus should remain on **high-revenue** areas like **elective admissions**, especially for **Cancer** and **Diabetes**.
- The sharp drop in **2024 ($1.5M**) indicates a possible issue such as **reduced patients**, **incomplete** data, or **operational** problems that need immediate **investigation**.
- **Hospitals** should take steps to increase **patient inflow** and improve **treatment** efficiency, as **abnormal** cases are slightly **higher**.
- Regular monitoring of **monthly trends** and taking early corrective **actions** will help maintain **consistent** revenue and performance.